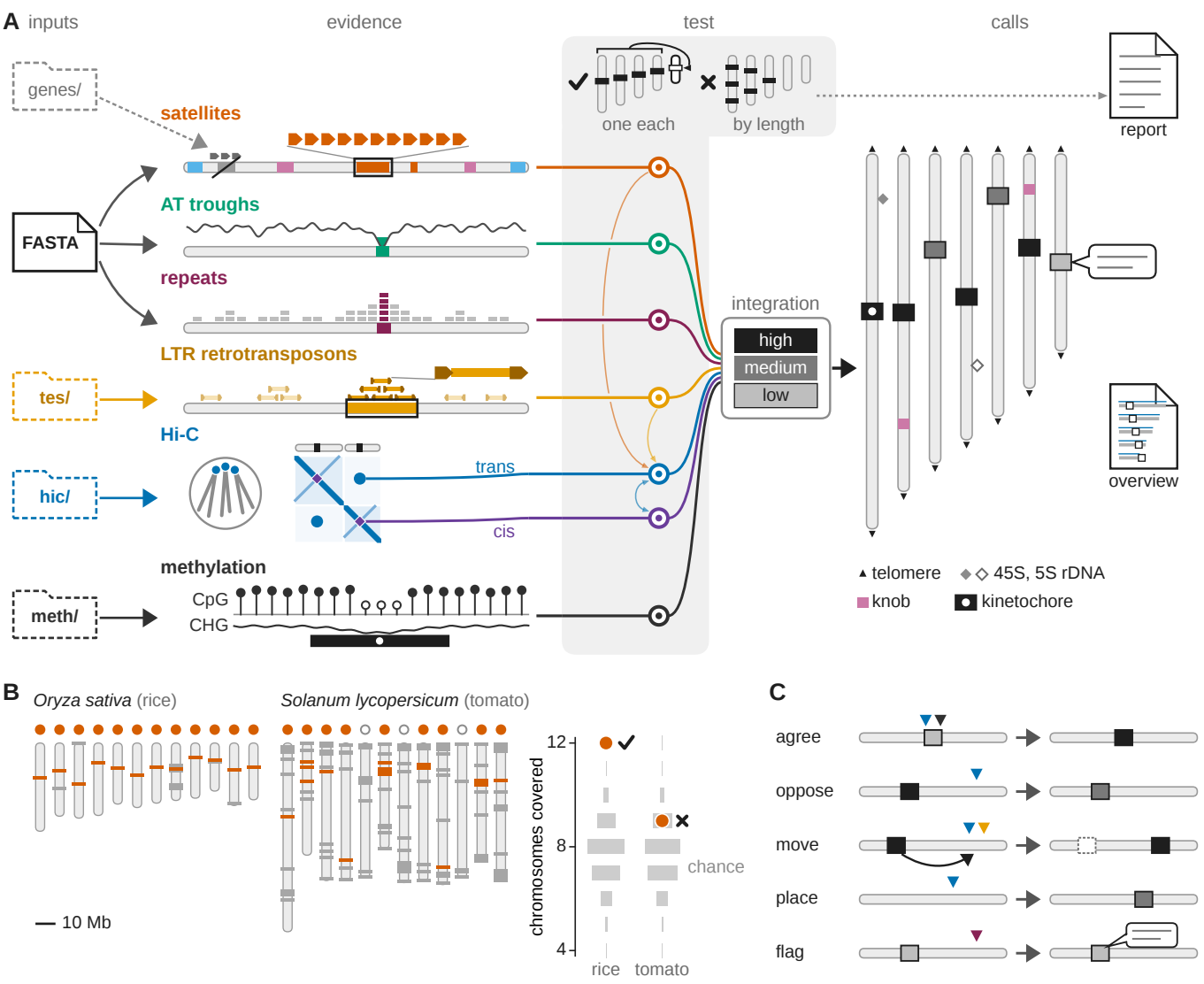

A
inputs
evidence
test
calls
genes/
FASTA
satellites
AT troughs
repeats
LTR retrotransposons
tes/
Hi-C
hic/
trans
cis
methylation
meth/
CpG
CHG
one each
by length
report
integration
high
medium
low
overview
telomere
45S, 5S rDNA
knob
kinetochore
B
Oryza sativa (rice)
Solanum lycopersicum (tomato)
10 Mb
chromosomes covered
12
8
4
chance
rice
tomato
C
agree
oppose
move
place
flag

A centromere occurs once per chromosome, so whatever marks centromeres falls once on each chromosome, while other repeats fall roughly in proportion to chromosome length. cenfuse lets each kind of evidence act only after it shows this pattern in the genome at hand, and grades every call.

A Six lines of evidence find centromere-like places on their own: three from the assembly alone (satellite arrays, AT-rich troughs, repeats) and three from optional inputs (dashed: retrotransposons, Hi-C read two ways, methylation). Each is tested in this genome, mostly by whether it marks one place per chromosome rather than places spread by length (gray column; each chromosome weighed against the others); a line that fails changes no call. Lines that pass, some tested against one another (thin links), give one call per chromosome, graded high, medium or low and explained in words (bubble).

B The test on real satellites: every cluster of the centromeric satellite families, to scale (drawn at least 1.5 Mb tall); orange, the 12 largest; filled dots, the chromosomes they reach. In rice they reach all 12 chromosomes, far more than the 8 expected by chance; in tomato they pile onto some chromosomes and miss others (9 of 12), as chance would, so its satellite calls stay low.

C Tested lines adjust the call from the sequence (left: before; right: after; triangles: where a tested line points, in its color). Each agreeing line raises the call one level, two at most. Hi-C, an LTR core or a CpG dip pointing elsewhere lowers it one level; strong evidence elsewhere moves it, a high call only with two lines (dashed: the old place, still reported). Where the sequence gives no call, Hi-C or an LTR core places one at medium, troughs or repeats alone at low, and a trough or repeat peak at a Hi-C call lifts it one level. A repeat line pointing away from a low call leaves it in place and flags it in words (bubble).

Data: docs/img/src/workflow_panelB.tsv (cenfuse 0.16.0 benchmark runs) · Script: docs/img/src/workflow.py · 2026-11-14